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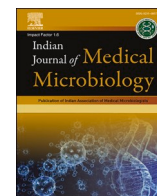


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Antimicrobial efficacy of eravacycline against emerging extensively drug-resistant (XDR) *Acinetobacter baumannii* isolates

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ABSTRACT

Purpose: Drug-resistant *Acinetobacter baumannii* is an emerging threat. This study has been conducted to observe the efficacy of eravacycline along with the RND-efflux pump system.

Methods: A cross-sectional study was done collecting 48 clinical isolates of *Acinetobacter baumannii*. MICs of 15 antibiotics were detected along with BMD of tigecycline and eravacycline. PCR products of drug-resistant regulatory genes were sequenced and analyzed.

Results: Of the total 48 Isolates, 35 (72.91%) were XDR and 13 (27.08%) were MDR. Out of all, 60.41% of isolates were found to be susceptible to eravacycline by BMD according to both FDA and EUCAST guidelines. A 2-fold decline of MIC_{50/90} was observed with the use of eravacycline compared to tigecycline. RND-efflux genes like AdeC in 30 (62.5%) isolates and Regulatory gene AdeS in 29 (60.41%) isolates were detected, explaining the existing resistance mechanism.

Conclusions: XDR *Acinetobacter* poses an escalating threat due to its resistance to multiple antibiotics, raising serious concerns in healthcare settings. Eravacycline is an encouraging new drug for empirical use in severe infection caused due to the same. Molecular investigation and strict antimicrobial stewardship should be followed to control the emergence, and a better understanding of mechanisms of resistance to prevent the spread of drug-resistant isolates.

1. Introduction

Acinetobacter baumannii is an opportunistic pathogen in humans causing various clinical manifestations that include nosocomial pneumonia, bacteremia secondary to progressed pneumonia, catheter-associated bacteremia, urinary tract infection, soft tissue infection and less commonly endocarditis, meningitis, osteomyelitis and, necrotizing fasciitis [1,2]. Globally, infection and outbreaks of *A. baumannii* are becoming more common and is in alarming situation [3]. Indeed, in the Asian subcontinent and Latin American countries *A. baumannii* infections are currently one of the top three most common causes of bacteremia and nosocomial pneumonia [1,4]. Infections of *A. baumannii* have been associated with the galloping frequencies of intensive care interventions such as ventilations, catheterizations and antibiotic therapy [5]. *A. baumannii* has been classified as a serious threat by World

Health Organization (WHO) as well as carbapenem-resistant *A. baumannii* as an urgent threat by Centre for Diseases Control (CDC) [6,7].

Throughout the world, emerging antimicrobial-resistant pathogen continues to pose a huge problem in the healthcare system and, at least in some part, is playing a key role in an initiative to develop new antimicrobials, including novel classes and novel agents within existing classes of antibiotics. *A. baumannii* has shown the highest rate of drug resistance of any gram-negative bacteria that leads to nosocomial infection [8,9] and there is quick emergence and evolution of drug resistance, Multidrug resistance (MDR), extensively drug resistance (XDR) and pan drug resistance (PDR) isolates. MDR was defined as acquired non-susceptibility to at least 1 agent in ≥3 antimicrobial categories, XDR was defined as non-susceptibility to at least 1 agent in all but ≤2 antimicrobial categories (i.e., bacterial isolates remain susceptible to

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only 1 or 2 categories) and PDR was defined as non-susceptibility to all agents in all antimicrobial categories. To ensure correct application of these definitions, bacterial isolates should be tested against all of the antimicrobial agents within the antimicrobial categories [10].

In *Acinetobacter baumannii*, various drug-resistant mechanisms exist, most common is an alteration of penicillin-binding proteins and chromosomal efflux pump system. Resistance Nodulation cell division (RND)-efflux pump is well studied and characterized among chromosomal efflux pump systems. In *A. baumannii*, expression of RND-efflux pump gene AdeABC is controlled by a two-components regulatory system AdeRS that forms a signal transduction pathway in bacteria. AdeS protein is a sensor kinase and has histidine kinase activity while the protein AdeR is a transcriptional regulator [11] which directly binds to the DNA to regulate the gene expression. Various studies are reporting the role of mutations in AdeS and AdeR in overexpression of AdeABC pump that imparts the emergence of Multidrug drug resistance (MDR) and extensive drug resistance (XDR) pathogens. The substitutions in AdeS are one of the most prevalent mutations responsible for Multidrug resistance in *A. baumannii* [12].

Extensively drug-resistant *Acinetobacter baumannii* has compromised the potency of most of the treatment regimens and is mainly dependent on colistin and tigecycline [13,14]. However, tigecycline resistance has emerged since its first application in 2005 [15,16] and the most common mechanism due to which drug resistance was conferred, was the over-expression of non-specific efflux pump and mutation in ribosomal protection protein (RPPs) [17]. Some studies have shown up to 50% resistance rate of tigecycline and 20% resistance to polymyxin [18,19]. Therapeutic failure of the commonly used antibiotics against MDR and XDR strains leads to increased patient fatality. It becomes imperative for us to look for an effective regimen against emerging drug-resistant pathogens.

An alternative therapy, new derivatives of tetracycline has emerged i.e. Eravacycline (TP-434) a novel fluorocycline. It belongs to the tetracycline class of antimicrobials and is currently being developed and used as an oral and intravenous medication for life-threatening infections caused due to antibiotic-resistant bacteria [20]. Eravacycline is a dynamic, mechanism-based bacterial ribosome inhibitor. However, it is least affected by efflux pumps specific to tetracycline, ribosomal protection, and inactivation. These factors make eravacycline a potential and prospective agent for the treatment of infection caused by multidrug-resistant organisms [21]. Like other tetracycline, eravacycline prevents the incorporation of amino acid residues into elongation peptide chains by reversibly binding to the bacterial 30s Ribosome disrupting bacterial protein synthesis. Eravacycline has shown a tenfold higher affinity for ribosomal binding which inhibited protein translation at a fold lower concentration than tetracycline. However, the continuous and rampant use of antibiotics paves the way for the development of drug-resistant pathogens. We have carried out this experiment to assess the efficacy of eravacycline in the emerging drug resistance *Acinetobacter baumannii*.

2. Materials and methods

The study was designed and conducted at the Department of Microbiology, All India Institute of Medical Sciences, New Delhi from October 2019 to April 2022. A total of 48 *Acinetobacter* species isolates were collected from various sterile body fluids i.e., CSF (Cerebrospinal fluid), Pleural fluid, Ascitic fluid, Peritoneal fluid and Endotracheal (ET) aspirates, were identified by Matrix-assisted laser desorption/ionization Time of flight (MALDI-TOF). Isolates were further processed for the determination of Minimum inhibitory concentration (MIC) for various antibiotics by VITEK-2 AST system (Biomerieux, France) and Broth Microdilution (BMD) for tigecycline and eravacycline. DNA was extracted from the isolates by QIAMP mini DNA extraction kit (Qiagen, USA).

2.1. Bacterial identification by MALDI-TOF

Sterile body fluids (Ascitic fluid, CSF, Peritoneal fluid, Pleural fluid and ET aspirates) were inoculated on Blood, Chocolate and Mac Conkey agar plates and were incubated at 37 °C for 24–48 h. A smear of isolates grown on culture plate was made on MALDI-TOF slide. 0.5 µl matrix (α -cyano- 4 hydroxycinnamic acid) was added to the smear and kept for 1 min at room temperature. Slide was kept in MALDI-TOF (VITEK-MS, Biomerieux, France) machine for acquisition and identification.

2.2. MIC determination methods

2.2.1. MIC determination by VITEK-2 AST system

Acinetobacter baumannii isolates were further processed for MIC determination of commonly used 15 antibiotics by VITEK-2 AST System (Biomerieux, France) using VITEK Card 281, following the standard operation procedure of the manufacturer. VITEK give results according to FDA cutoffs for tigecycline in case of *Acinetobacter* [Susceptible (MIC) ≤ 2 mg/L, Intermediate (MIC) > 2 and < 8 mg/L and Resistant (MIC) ≥ 8 mg/L].

2.2.2. MIC determination by broth microdilution (BMD) for tigecycline and eravacycline

Broth microdilution is used to measure *in vitro* activity of antibiotics against bacterial isolates. A sterile 96-well plastic tray containing different concentrations of antibiotic (five to eight concentrations) was inoculated with standardized test bacterial suspension in double strength CAMHB (Cation-adjusted Mueller Hinton broth). Plate was incubated at 30 °C for 18–24 h. MIC (Minimum inhibition concentration) was determined by observing the lowest concentration of antibiotics that inhibit the visible growth of bacteria. Broth micro-dilution (BMD) was performed to determine the MIC_{50/90} for eravacycline v/s tigecycline. MIC₅₀ is the minimum inhibitory concentration required to inhibit the growth of 50% of the organism and MIC₉₀ is the minimum inhibitory concentration required to inhibit the growth of 90% of the organism.

For *Acinetobacter* there are no breakpoints defined for tigecycline and eravacycline by European Committee on Antimicrobial Susceptibility Testing (EUCAST 2020) and Food and Drug Administration (FDA 2020). The Clinical & Laboratory Standards Institute (CLSI 2020) does not publish susceptible breakpoints for tigecycline & eravacycline [22]. EUCAST, 2020 reports insufficient evidence (IE) for tigecycline & eravacycline breakpoints for *Acinetobacter* species. “IE” indicates that there is insufficient evidence that the organism or group is a good target for therapy with the agent. A MIC with a comment but without an accompanying S, I or R categorization may be reported. Based on the literature, we have used the *Enterobacterales* breakpoint tables for both eravacycline and tigecycline.

The U.S. Food and Drug Administration (FDA 2020) recommendation for.

a Tigecycline susceptibility breakpoints of *Enterobacterales* were used as the MIC interpretation criteria for *Acinetobacter* spp. are:

Susceptible (MIC) ≤ 2 mg/L
Intermediate (MIC) > 2 and < 8 mg/L
Resistant (MIC) ≥ 8 mg/L

b Eravacycline breakpoints of *Enterobacterales* are:

Susceptible ≤ 0.5 mg/L
Resistant > 0.5 mg/L

According to the European Committee on Antimicrobial Susceptibility Testing (EUCAST, 2020) breakpoint tables for interpretation of MICs and zone diameters for *Acinetobacter* spp. Indicates insufficient evidence for tigecycline & eravacycline.

The EUCAST 2020 recommendation for.

- a Tigecycline breakpoints of *Enterobacterales* are:
 Susceptible ≤ 0.5 mg/L
 Resistant > 0.5 mg/L
- b Eravacycline breakpoints of *Enterobacterales* are:
 Susceptible ≤ 0.5 mg/L
 Resistant > 0.5 mg/L

Based on the literature we used the *Enterobacterales* breakpoint tables for both tigecycline & eravacycline MIC breakpoints, which have been set at 0.5 mg/L by EUCAST 2020.

For BMD of tigecycline (Sigma Aldrich, USA), *Escherichia coli* ATCC 25922 (Ref MIC value 0.03–0.25 μ g/ml) and *Enterococcus faecalis* ATCC (Ref MIC value 0.03–0.12 μ g/ml) were used as control strains. Similarly, for eravacycline (Adooq Bioscience), control strains *Escherichia coli* ATCC 25922 (Reference MIC value 0.016–0.004 μ g/ml) and *Pseudomonas aeruginosa* ATCC 27853 (Reference MIC value 2–8 μ g/ml) was used.

2.3. PCR of RND efflux pump genes and sequencing

PCR was performed for the detection of RND efflux pump genes AdeA, AdeB and AdeC, AdeR and AdeS. The cycling condition used for PCR was Initial denaturation at 94 °C for 10 min, denaturation at 94 °C for 1 min, annealing at 56 °C for 30 s, extension at 72 °C for 1 min and final extension at 72 °C for 7 min. Total number of cycles were 35. Representative PCR products of RND efflux pump genes AdeABC and Regulatory gene AdeRS were sequenced by Sanger sequencing method. Sequences were analyzed by Genedoc software version 2.7.

3. Statistical analysis

Data were analyzed using non-parametric test by Fischer exact test and correlation between MDR/ XDR and RND-efflux pump gene AdeABC and Regulator gene AdeRS were sought.

4. Results

A total of 48 antimicrobial-resistant isolates of *Acinetobacter baumannii* were collected from different sterile body fluids irrespective of age group. Mean age of the patients was 40.48 ± 20.62 years. Maximum number of *A. baumannii* was detected from CSF 14 (29.16%), followed by Pleural fluid 13 (27.08%), Ascitic fluid 8 (16.66%), ET aspirates 8 (16.66%) and Peritoneal fluids 5 (10.41%). Overall, most of the strains were highly drug-resistant to common drugs. 35 (72.91%) isolates were XDR, only sensitive to ≤ 2 groups of drugs following the definition of XDR for *Acinetobacter baumannii* has been used [10]. Isolates were highly resistant to β -lactam drugs, cephalosporin group, carbapenem and β -lactamase inhibitor combination. However, all isolates were resistant to trimethoprim-sulfamethoxazole. Among all groups of antibiotics, the most effective were tigecycline (47.91%), cefepime (52.08%) and doripenem (56.25%), detected by VITEK-2 AST system as shown in Table 1. XDR isolates were mainly sensitive to last-resort drugs like tigecycline and colistin.

MIC of tigecycline and eravacycline was detected by BMD, and MIC₅₀/MIC₉₀ were calculated as shown in Table 2. Out of the total, 23 (47.91%) *A. baumannii* isolates had tigecycline MIC ≥ 0.5 μ g/ml, which falls in the resistant category for *Enterobacterales* as per EUCAST 2020 guideline [23]. 8 (16.67%) *A. baumannii* isolates showed a higher level of resistance to tigecycline (MIC value ≥ 2 μ g/ml) and 3 (6.25%) strains had a MIC value of 4 μ g/ml making it an intermediate category as per the FDA breakpoint. For eravacycline, 29 (60.41%) isolates were susceptible by both FDA and EUCAST guidelines. A 2-fold decline of MIC₅₀/90 was observed with the use of eravacycline.

RND-efflux pump genes AdeABC and regulatory genes AdeRS have been detected in all the drug-resistant isolates of *Acinetobacter baumannii*. Out of 6 strains with eravacycline MIC value 2 μ g/ml, 4

Table 1

Interpretation of antibiotics used against *Acinetobacter baumannii* by VITEK-2 system.

Antibiotics	Susceptible	Intermediate	Resistance
Ticarcillin/ Clavulanic acid	41.66%	0	58.34%
Piperacillin-tazobactam	29.16%	0	70.84%
Ceftazidime	31.25%	0	68.75%
Cefoparazone-sulbactam	22.91%	0	77.09%
Cefepime	52.08%	0	47.92%
Doripenem	56.25%	0	43.75%
Imipenem	20.83%	0	79.17%
Meropenem	33.33%	0	66.67%
Amikacin	29.16%	0	70.84%
Gentamicin	25.00%	0	75.00%
Ciprofloxacin	27.08%	0	72.92%
Levofloxacin	25.00%	22.91%	52.09 %
Tigecycline	47.91%	33.33%	14.58%
Colistin	0 %	89.60%	10.40%
Trimethoprim-sulfamethoxazole	0	0	100%

(66.67%) isolates had RND efflux pump genes AdeABC and AdeRS. However, in one isolate only AdeA was detected and in one isolate only Regulatory genes AdeR and AdeS were detected. In *Acinetobacter baumannii*, RND-efflux pumps genes AdeABC confers resistance to various antibiotics. AdeABC and Regulatory system genes AdeRS genes detected by PCR are shown in Fig. 1.

Out of 48 *Acinetobacter* isolates, 38 (79.17%) isolates were positive for efflux pump genes. AdeC genes were detected in 29 (60.41%) isolates followed by AdeB genes in 27 (56.25%) isolates. Similarly, the Regulatory gene of RND efflux pump AdeS was detected in 30 (60.5%) isolates. Sequencing was performed for fourteen representative positive samples for RND efflux pump genes AdeABC and AdeRS. Sequences were analyzed by Basic Local Alignment Search Tool (BLAST) and it revealed mutation in efflux pump regulatory gene AdeS at position L172 P (Accession no. of AdeS gene was MT912027). Sequences were submitted to NCBI Gene Bank with accession no. MT912014, MT912015, MT912016, MT912017, MT912018, MT912019, MT912020, MT912021, MT912022, MT9120123, MT912024, MT912025, MT912026 and MT912027.

Data were analyzed using non-parametric test by Fischer exact test and correlation between MDR/ XDR and RND-efflux pump gene AdeABC and Regulator gene AdeRS were sought. P value < 0.005 was statistically significant. P value of AdeB 0.004, AdeR 0.005 and AdeS 0.000 was found.

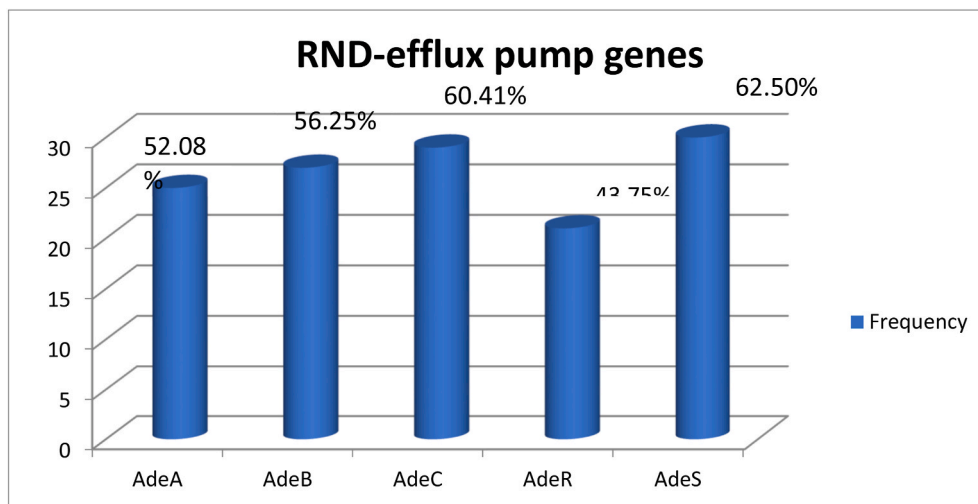
5. Discussion

To our knowledge, this was the first study from India to study the efficacy of eravacycline in antimicrobial-resistant *Acinetobacter baumannii* along with other phenotypic and genotypic profiles. Multidrug-resistant *A. baumannii* is a well-established pathogen in many medical centers. However, options for therapy for this pathogen are also quite limited and often involve colistin and/or tigecycline and are used as a last-line drug. In this study, tigecycline showed 14.58% and 47.91% resistance by VITEK-2 and BMD respectively which is in concordance with the finding of Qureshi Z A et al. and Tsioutis C et al. Cutoffs for the “resistant” category have higher breakpoints in VITEK-2 system (following FDA guidelines) which makes the actual chunk of resistant isolates to be categorized as sensitive or intermediate. In the resistant isolates, efflux pump genes were found to be the most cardinal mechanism of developing drug resistance. 38 (79.17%) isolates were positive for efflux pump genes. The lower MIC value of eravacycline may be due to its newness and use for a lesser duration of time as compared to other drugs.

Eravacycline demonstrated activity against these isolates and was 2-fold more active than tigecycline. In our study, increased MIC values to eravacycline (reaching 2–4 μ g/ml) were associated with increased

Table 2MIC values of Tigecycline and Eravacycline for *Acinetobacter baumannii* and MIC₅₀ and MIC₉₀ values as per EUCAST.

Antimicrobial agents	MIC value of antibiotics											% of Sensitive, Intermediate & Resistant		
	0.125	0.25	0.5	1	2	4	8	16	MIC ₅₀	MIC ₉₀	MIC Range	%S	%I	%R
Tigecycline	5	17	3	15	5	3	-	-	0.5	4	0.125–4	52.08	-	47.91
Eravacycline	8	16	5	13	6	-	-	-	0.25	2	0.125–4	61.41	-	38.59

**Fig. 1.** Bar diagram showing the distribution of RND-efflux pump and Regulatory genes.

expression of the native efflux pump gene AdeABC. Increased expression of this RND-efflux system has also been associated with reduced activity of tigecycline [24]. Eravacycline has a unique chemical modification at C-9 and C-7 of the core of tetracycline due to which it confers potent, broad-spectrum antibacterial activity, especially against multidrug-resistant, extensively drug-resistant and difficult-to-treat pathogens. Previously, the activity of eravacycline was shown to be affected minimally by tetracycline-specific efflux and ribosome protection and inactivation [25].

Due to the evolution and emergence of the isolates expressing multiple antibiotic-resistant mechanisms, it is becoming more difficult to treat life-threatening infections caused due to Gram-negative bacteria i. e., *A. baumannii* and *Enterobacteriales* that produce ESBL [26]. In hospital and community settings multidrug-resistant organisms are of particular public health concern [27]. In clinical use, tigecycline long approved tetracycline-class antibiotic; however, eravacycline is differentiated from tigecycline, by its in situ superior activity across multiple organisms, especially multidrug-resistant *Acinetobacter* spp. and ESBL-producing *Enterobacteriales*, as well as by its encouraging pharmacokinetics, potential for oral dosing and tolerability [28,29].

Mutation in RND-efflux pump regulatory gene AdeS at position L172 P was observed in one isolate after sequence analysis. Point mutations in a protein are known to induce conformational changes in the three-dimensional structure of the protein and these altered conformations may be responsible for ceased or enhanced activity of the mutant protein [30]. The mutation may be responsible for the emergence of drug resistance isolates. Hence, along with regular monitoring of antibiotic susceptibility patterns and MIC of the different isolates, molecular diagnosis should also be implemented to find out the genotypic changes. Regular, phenotypic and genotypic surveillance helps to prevent from development multidrug multidrug-resistant pathogens as well as alarm the system to get ready with the new arsenals in the case of the emergence of drug-resistant organisms.

6. Conclusion

Eravacycline is an encouraging new drug for empirical use in severe infection caused due to extensively drug-resistant *A. baumannii*. It can potentially become a useful substitution and addition to the existing armamentarium of antibiotics that can treat this problem effectively. Mutation plays a very crucial role in developing drug resistance. The presence of a single mutation at Leu172Pro in the sensor domain AdeS indicates the significance of mutation. Though, structural and biochemical studies need to be carried out to get deeper insights into the role of different mutations in the overexpression of RND efflux pump genes AdeABC. In addition, antimicrobial stewardship should be strictly implemented to prevent from spread of resistant isolates in hospitals, especially in rural areas where rampant use of antibiotics is not uncommon.

Ethics statement

Institute Ethics Committee, AIIMS, New Delhi has approved the work with reference no- IECPG-773/30.01.2020.

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CRedit authorship contribution statement

Hitender Gautam: Writing – original draft, (lead), Data curation, (Lead), Writing – review & editing, (equal), Conceptualization, (lead), Supervision, (lead), Resources, (lead). **Shahid Raza:** Formal analysis, (lead), Methodology, (lead). **Jaya Biswas:** Formal analysis, (lead), Methodology, (lead). **Sarita Mohapatra:** Writing – review & editing, (equal). **Seema Sood:** Writing – review & editing, (equal). **Benu Dhanwan:** Writing – review & editing, (equal). **Arti Kapil:** Supervision, (equal). **Bimal K. Das:** Supervision, (equal), Resources, (equal).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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